Lecture 1
Solutions

Homework 1: Your first repository

Solutions to Homework 1

Run these from the course folder `LearningGit/`. Your prompt and the exact path will differ.

```
$ cd practice
$ mkdir hw1
$ cd hw1
$ git init -b main
Initialized empty Git repository in .../practice/hw1/.git/
$ echo "Hello Git" > hello.txt
$ git config --local user.name "Practice User"
$ git config --local user.email "practice@example.com"
$ git status > status.txt
$ echo ".git" > answer.txt
```

1.1. `mkdir hw1` creates the folder and `cd hw1` enters it (do this from inside `practice/`).

1.2. `git init` creates the repository. The branch name comes from the global setting you made in Homework 0; `git init -b main` states it explicitly (Git 2.28 or newer), so it is the safe choice. If you forgot and your branch has another name (for example `master`), rename it with `git branch -m main`.

1.3. `echo "Hello Git" > hello.txt` writes one line. Check with `cat hello.txt`. Note that `>` *replaces* the file each time; `>>` would append.

1.4. `--local` is the default for `git config` inside a repository, so you could omit it, but writing it makes the intent clear. Verify where the values come from, and that your global ones are unchanged:

```
$ git config --show-origin user.name
file:.git/config        Practice User
$ git config --global user.name
Your Name
```

1.5. After the redirect, `status.txt` should look like this (it lists itself as untracked too, because the file already exists when Git runs):

```
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        hello.txt
        status.txt

nothing added to commit but untracked files present (use "git add" to track)
```

1.6. The folder is `.git`. The file may contain `.git` or `.git/`.

Common mistakes.

- Running `git init` in the wrong folder (check with `pwd`). Do *not* delete anything until you are sure: `rm -r .git` is only safe in a folder you created for this exercise and just ran `git init` in by mistake. Never run it inside `LearningGit/` itself, because that would erase the course repository's history.
- Forgetting that `git init` is needed inside `practice/hw1`. Without it, `git status` would talk about the *course* repository above it.
- Typing the `$` at the start of a command.
- Using `--global` instead of `--local` in 1.4. That changes your name for every project. Undo it by setting your real name and email again with `git config --global`.